Table 8.3: Thermal conductivities (in W m^{-1} K^{-1}) of some materials at 1 *atm*

Material	Thermal Conductivity	Material	Thermal Conductivity
Diamond	2300	Water	0.56
Silver	420	Human tissue	0.2
Copper	380	Wood	0.17
Aluminum	200	Helium	0.152
Steel	40	Cork	0.042
Glass	0.84	Air	0.023
Brick	0.84		
Ice	2		

Note

Steady state:
The state at which temperature attains constant value everywhere and there is no further transfer of heat anywhere is called steady state.

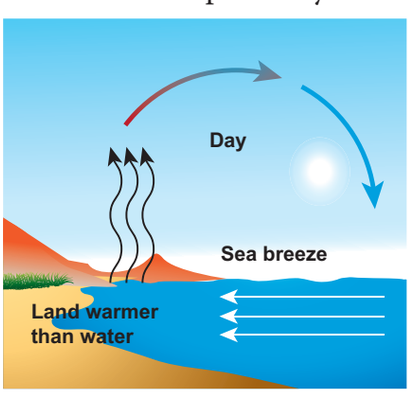
Thermal conductivity depends on the nature of the material. For example silver and aluminum have high thermal conductivities. So they are used to make cooking vessels.

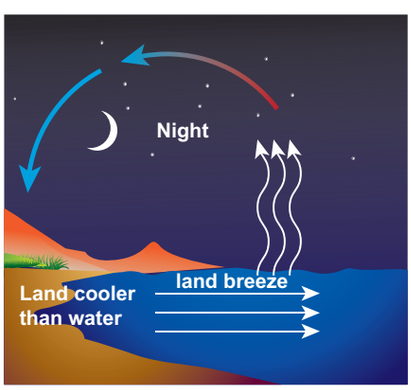
Convection

Convection is the process in which heat transfer is by actual movement of molecules in fluids such as liquids and gases. In convection, molecules move freely from one place to another. It happens naturally or forcefully.

DO YOU KNOW?

During the day, sun rays warm up the land more quickly than sea water. It is because land has less specific heat capacity than water. As a result the air above the land becomes less dense and rises. At the same time the cooler air above the sea flows to land and it is called 'sea breeze'. During the night time the land gets cooled faster than sea due to the same reason (specific heat). The air molecules above sea are warmer than air molecules above the land. So air molecules above the sea are replaced by cooler air molecules from the land. It is called 'land breeze'.





UNIT-8(XI-Physics_Vol-2).indd 108 02-01-2023 15:25:34